

Assignment 4: Student Exam Score Prediction Summary Report

1. Project Overview

This project focuses on predicting student exam scores by various study habits and personal factors.

2. Data Processing & Personalization

The data preparation phase involved tailoring the dataset to a specific student demographic and ensuring high data quality through rigorous cleaning procedures.

- **Data Personalization:** To personalize the analysis, the dataset was filtered exclusively for male students. Additionally, a specific study-hour threshold of 1 hour per day was applied, determined by the student's roll number.
- **Data Cleaning:** The dataset underwent a comprehensive cleaning process:
 - Verified the absence of missing values and duplicates.
 - Addressed potential outliers to maintain model stability.
 - Imputed categorical missing values using the **Mode** to preserve the statistical integrity of the categorical features.

3. Exploratory Data Analysis (EDA)

Exploratory analysis provided critical insights into the underlying patterns and relationships within the data.

- **Score Distribution:** The analysis revealed that exam scores are widely distributed, with a notable clustering in the middle-to-high performance range.

- **Study Habits Correlation:** A strong positive relationship was observed between the number of study hours and final exam scores, reinforcing the importance of consistent academic engagement.
- **Lifestyle Impact:**
 - **Part-time Jobs:** Evaluated the relationship between part-time employment and academic outcomes.
 - **Sleep Patterns:** Analyzed average scores across different sleep hour buckets to understand the impact of rest on cognitive performance.
- **Correlation Heatmap:** The heatmap highlighted significant positive correlations between **study hours**, **attendance**, and **exam performance**, identifying these as the most influential predictors.

4. Modeling & Evaluation

A predictive framework was established using a Linear Regression approach to quantify the impact of student behaviors on their scores.

- **Model Architecture:** A Linear Regression model was trained using an 80/20 train-test split to ensure robust evaluation.
- **Refinement:** A comparative analysis was conducted between the full model and a streamlined version focusing on core features: study hours, attendance, sleep, and mental health.

5. Conclusion

The analytical process successfully identified **study hours** and **attendance** as the primary positive drivers of exam performance. The implemented model achieved an R^2 score of 0.70, satisfying the assignment's passing criteria and demonstrating sufficient predictive power for academic assessment purposes.